

SPECIFICATIONFOR APPROVAL

承 认 书

客户名称 (CUSTOMER): 温州远帆科技

产品名称 (PRODUCT ITEM): 螺纹温度传感器

文件编号 (FILE NO) : DCSL2025040101

编日期 (DATE): 2025 年 4 月 1 日

客户确认(CUSTOMER CONFIRM)		批准 (APPROVE)
注: 承认书内容确认后,请盖骑缝章		
客户确认栏(签字并盖章)		审核(CHECK)

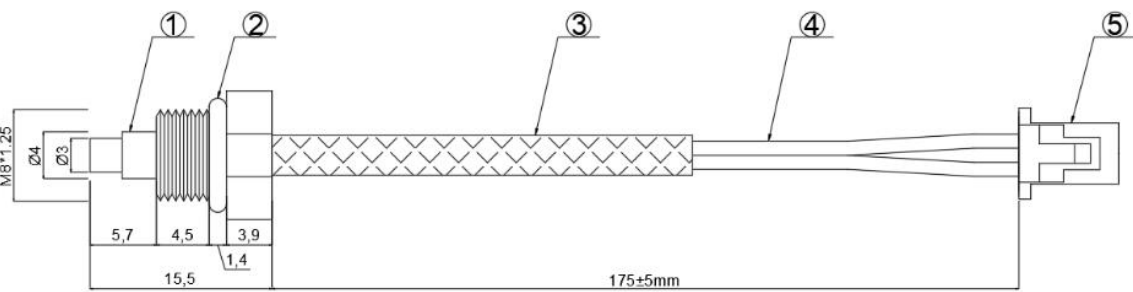
Zhaoqing duancheng electronic technology co.,ltd

肇庆市端诚电子科技有限公司

规格型(PART NO)	DCSL-503F3950-170CB
文件编号(FILE NO)	DCSL2025040101

产品规格书

1.外形尺寸



2.外形规格

序号	材料名称	规格/型号	数量	备注
1	外壳	SL11*14.5+M8 螺纹+硅胶圈	1pcs	
2	电阻	R25=50K B25/50=3950 ±1%	1pcs	
3	套管	3.0*140mm 黄腊管	1pcs	
4	电线	26#2651 黑色	1pcs	
5	端子连接器	XHB 白色	1pcs	

3.电气性能

序号	项目	符号	测试条件	最小值	正常值	最大值	单位
3-1	25℃的电阻值	R_{25}	$T_a=25\pm0.05^{\circ}\text{C}$ $P_t\leq0.1\text{mw}$	49.5	50	50.5	$\text{k}\Omega$
3-2	50℃的电阻值	R_{50}	$T_a=50\pm0.05^{\circ}\text{C}$ $P_t\leq0.1\text{mw}$	17.58	17.95	18.30	$\text{k}\Omega$
3-3	B 值	$B_{(25/50)}$	$B = LN \frac{R_{T1}}{R_{T2}} / (\frac{1}{T1} - \frac{1}{T2})$	3910.5	3950	3989.5	k

3-4	耗散系数	σ	Ta=25±0.5℃	5	/	/	mw/℃
3-5	时间常数	τ	Ta=25±0.5℃	/	/	10	sec
3-6	绝缘电阻	/	500Vdc	100	/	/	
3-7	耐压测试	/	1500V AC, I=0.5mA	60	/	/	s
3-8	使用温度范围	/		-30	/	+105	℃

4.可靠性试验

序号	项目	技术要求	测试条件及方法
4-1	高温试验	$\Delta R/R_{25} \leq \pm 3\%$	100±5℃,通电 1000±24h, DC0.2mA
4-2	低温试验	$\Delta R/R_{25} \leq \pm 3\%$	-30±5℃,通电 1000±24h, DC0.2mA
4-3	温度冷热循环试验	$\Delta R/R_{25} \leq \pm 3\%$	-2℃×30→常温×10min→100℃水中×3min→常温×10min, 来回共 10 循环周期
4-4	耐潮湿试验	$\Delta R/R_{25} \leq \pm 3\%$	55±2℃,90%-95%RH 环境下放置 1000±24h

5.机械性能

序号	项目	技术要求	测试条件及方法
5-1	拉力测试	树脂、不变形、无损伤	加载 10n(3F)1 分钟
		端子、不变形、无损伤	加载 10n(3F)1 分钟
5-2	落地测试	无可见性损伤	在 1 米的高度,让产品做自由落体运动,下落到 10mm 厚的橡木板上, 5 次

6. 贮存方法

6.1 允许用任何方法运输,但要避免雨、雪的直接或间接的淋袭和机械损伤

6.2 产品应贮存环境在温度为 10℃/+40℃,相对湿度不大于 80%,周围环境不应有酸性、碱性物质及腐蚀气体或辐射源.

R-T 表 R25=50K Ω $\pm$ 1% B25/50=3950K $\pm$ 1%

Temp. (deg. C)	Rmax (k Ohms)	Rnor (k Ohms)	Rmin (k Ohms)	DR		dT (deg. C)	
-30	929.3464	893.9899	859.8925	4%	-4%	0.7	-0.7
-29	872.4685	839.8008	808.2755	4%	-4%	0.6	-0.6
-28	819.4456	789.2531	760.0970	4%	-4%	0.6	-0.6
-27	769.9924	742.0792	715.1064	4%	-4%	0.6	-0.6
-26	723.8464	698.0331	673.0729	4%	-4%	0.6	-0.6
-25	680.7657	656.8877	633.7838	4%	-4%	0.6	-0.6
-24	640.5280	618.4343	597.0428	4%	-3%	0.6	-0.6
-23	602.9280	582.4799	562.6691	4%	-3%	0.6	-0.6
-22	567.7766	548.8470	530.4955	3%	-3%	0.6	-0.6
-21	534.8996	517.3716	500.3679	3%	-3%	0.6	-0.6
-20	504.1358	487.9019	472.1436	3%	-3%	0.6	-0.6
-19	475.3366	460.2981	445.6908	3%	-3%	0.6	-0.6
-18	448.3647	434.4308	420.8877	3%	-3%	0.6	-0.6
-17	423.0932	410.1801	397.6214	3%	-3%	0.6	-0.6
-16	399.4046	387.4355	375.7875	3%	-3%	0.6	-0.6
-15	377.1905	366.0945	355.2893	3%	-3%	0.6	-0.6
-14	356.3503	346.0619	336.0370	3%	-3%	0.5	-0.5
-13	336.7907	327.2499	317.9477	3%	-3%	0.5	-0.5
-12	318.4257	309.5770	300.9442	3%	-3%	0.5	-0.5
-11	301.1751	292.9673	284.9547	3%	-3%	0.5	-0.5
-10	284.9649	277.3508	269.9131	3%	-3%	0.5	-0.5
-9	269.7263	262.6622	255.7576	3%	-3%	0.5	-0.5
-8	255.3953	248.8411	242.4308	3%	-3%	0.5	-0.5
-7	241.9127	235.8311	229.8793	3%	-3%	0.5	-0.5
-6	229.2233	223.5799	218.0537	3%	-2%	0.5	-0.5
-5	217.2760	212.0390	206.9077	2%	-2%	0.5	-0.5
-4	206.0230	201.1632	196.3984	2%	-2%	0.5	-0.5
-3	195.4201	190.9102	186.4858	2%	-2%	0.5	-0.5
-2	185.4259	181.2409	177.1326	2%	-2%	0.5	-0.5
-1	176.0022	172.1188	168.3042	2%	-2%	0.5	-0.5
0	167.1132	163.5097	159.9680	2%	-2%	0.4	-0.4
1	158.7254	155.3820	152.0938	2%	-2%	0.4	-0.4
2	150.8079	147.7061	144.6536	2%	-2%	0.4	-0.4
3	143.3316	140.4542	137.6209	2%	-2%	0.4	-0.4
4	136.2695	133.6007	130.9711	2%	-2%	0.4	-0.4
5	129.5963	127.1214	124.6813	2%	-2%	0.4	-0.4
6	123.2885	120.9939	118.7300	2%	-2%	0.4	-0.4
7	117.3241	115.1970	113.0972	2%	-2%	0.4	-0.4
8	111.6825	109.7112	107.7639	2%	-2%	0.4	-0.4

9	106.3445	104.5181	102.7127	2%	-2%	0.4	-0.4
10	101.2920	99.6004	97.9272	2%	-2%	0.4	-0.4
11	96.5083	94.9420	93.3918	2%	-2%	0.4	-0.4
12	91.9776	90.5280	89.0922	2%	-2%	0.3	-0.3
13	87.6852	86.3440	85.0149	2%	-2%	0.3	-0.3
14	83.6172	82.3770	81.1471	2%	-1%	0.3	-0.3
15	79.7608	78.6145	77.4770	1%	-1%	0.3	-0.3
16	76.1037	75.0449	73.9934	1%	-1%	0.3	-0.3
17	72.6347	71.6572	70.6858	1%	-1%	0.3	-0.3
18	69.3430	68.4413	67.5445	1%	-1%	0.3	-0.3
19	66.2187	65.3874	64.5601	1%	-1%	0.3	-0.3
20	63.2524	62.4867	61.7241	1%	-1%	0.3	-0.3
21	60.4353	59.7306	59.0282	1%	-1%	0.3	-0.3
22	57.7590	57.1111	56.4649	1%	-1%	0.3	-0.3
23	55.2158	54.6208	54.0268	1%	-1%	0.3	-0.3
24	52.7985	52.2527	51.7073	1%	-1%	0.2	-0.2
25	50.5000	50.0000	49.5000	1%	-1%	0.2	-0.2
26	48.3564	47.8566	47.3573	1%	-1%	0.2	-0.2
27	46.3153	45.8167	45.3189	1%	-1%	0.2	-0.2
28	44.3713	43.8746	43.3791	1%	-1%	0.3	-0.3
29	42.5193	42.0252	41.5328	1%	-1%	0.3	-0.3
30	40.7545	40.2637	39.7748	1%	-1%	0.3	-0.3
31	39.0723	38.5853	38.1006	1%	-1%	0.3	-0.3
32	37.4685	36.9858	36.5058	1%	-1%	0.3	-0.3
33	35.9390	35.4611	34.9861	1%	-1%	0.3	-0.3
34	34.4799	34.0072	33.5376	1%	-1%	0.3	-0.3
35	33.0878	32.6205	32.1567	1%	-1%	0.3	-0.3
36	31.7591	31.2977	30.8399	1%	-1%	0.3	-0.3
37	30.4907	30.0354	29.5838	2%	-2%	0.4	-0.4
38	29.2796	28.8305	28.3855	2%	-2%	0.4	-0.4
39	28.1228	27.6802	27.2418	2%	-2%	0.4	-0.4
40	27.0177	26.5818	26.1502	2%	-2%	0.4	-0.4
41	25.9618	25.5325	25.1079	2%	-2%	0.4	-0.4
42	24.9525	24.5301	24.1125	2%	-2%	0.4	-0.4
43	23.9876	23.5722	23.1616	2%	-2%	0.4	-0.4
44	23.0649	22.6565	22.2531	2%	-2%	0.4	-0.4
45	22.1824	21.7811	21.3848	2%	-2%	0.5	-0.5
46	21.3382	20.9439	20.5548	2%	-2%	0.5	-0.5
47	20.5303	20.1431	19.7612	2%	-2%	0.5	-0.5
48	19.7571	19.3770	19.0022	2%	-2%	0.5	-0.5
49	19.0169	18.6438	18.2762	2%	-2%	0.5	-0.5

50	18.3082	17.9421	17.5815	2%	-2%	0.5	-0.5
51	17.6294	17.2702	16.9167	2%	-2%	0.5	-0.5
52	16.9791	16.6269	16.2804	2%	-2%	0.5	-0.5
53	16.3560	16.0107	15.6711	2%	-2%	0.6	-0.6
54	15.7589	15.4204	15.0877	2%	-2%	0.6	-0.6
55	15.1865	14.8548	14.5288	2%	-2%	0.6	-0.6
56	14.6378	14.3127	13.9934	2%	-2%	0.6	-0.6
57	14.1115	13.7930	13.4803	2%	-2%	0.6	-0.6
58	13.6067	13.2947	12.9886	2%	-2%	0.6	-0.6
59	13.1225	12.8169	12.5172	2%	-2%	0.6	-0.6
60	12.6578	12.3585	12.0651	2%	-2%	0.7	-0.7
61	12.2118	11.9188	11.6316	2%	-2%	0.7	-0.7
62	11.7837	11.4968	11.2158	2%	-2%	0.7	-0.7
63	11.3727	11.0918	10.8168	3%	-2%	0.7	-0.7
64	10.9780	10.7030	10.4339	3%	-3%	0.7	-0.7
65	10.5989	10.3297	10.0664	3%	-3%	0.7	-0.7
66	10.2346	9.9712	9.7136	3%	-3%	0.7	-0.7
67	9.8846	9.6269	9.3749	3%	-3%	0.7	-0.7
68	9.5483	9.2960	9.0495	3%	-3%	0.8	-0.8
69	9.2250	8.9781	8.7370	3%	-3%	0.8	-0.8
70	8.9141	8.6726	8.4368	3%	-3%	0.8	-0.8
71	8.6152	8.3789	8.1482	3%	-3%	0.8	-0.8
72	8.3277	8.0965	7.8709	3%	-3%	0.8	-0.8
73	8.0512	7.8250	7.6044	3%	-3%	0.8	-0.8
74	7.7851	7.5638	7.3481	3%	-3%	0.8	-0.8
75	7.5290	7.3126	7.1016	3%	-3%	0.9	-0.9
76	7.2826	7.0708	6.8645	3%	-3%	0.9	-0.9
77	7.0454	6.8382	6.6365	3%	-3%	0.9	-0.9
78	6.8170	6.6143	6.4170	3%	-3%	0.9	-0.9
79	6.5970	6.3988	6.2059	3%	-3%	0.9	-0.9
80	6.3852	6.1913	6.0026	3%	-3%	0.9	-0.9
81	6.1811	5.9914	5.8070	3%	-3%	0.9	-0.9
82	5.9845	5.7989	5.6186	3%	-3%	1	-1
83	5.7950	5.6135	5.4372	3%	-3%	1	-1
84	5.6124	5.4348	5.2624	3%	-3%	1	-1
85	5.4363	5.2627	5.0941	3%	-3%	1	-1
86	5.2666	5.0968	4.9319	3%	-3%	1	-1
87	5.1030	4.9368	4.7756	3%	-3%	1	-1
88	4.9451	4.7826	4.6249	3%	-3%	1.1	-1.1
89	4.7929	4.6339	4.4797	3%	-3%	1.1	-1.1
90	4.6460	4.4905	4.3397	3%	-3%	1.1	-1.1

91	4.5043	4.3521	4.2047	3%	-3%	1.1	-1.1
92	4.3675	4.2186	4.0745	4%	-3%	1.1	-1.1
93	4.2355	4.0899	3.9489	4%	-3%	1.1	-1.1
94	4.1080	3.9656	3.8277	4%	-3%	1.1	-1.1
95	3.9850	3.8456	3.7108	4%	-4%	1.2	-1.2
96	3.8662	3.7298	3.5979	4%	-4%	1.2	-1.2
97	3.7514	3.6180	3.4890	4%	-4%	1.2	-1.2
98	3.6406	3.5100	3.3839	4%	-4%	1.2	-1.2
99	3.5335	3.4058	3.2823	4%	-4%	1.2	-1.2
100	3.4300	3.3051	3.1843	4%	-4%	1.2	-1.2
101	3.3300	3.2077	3.0896	4%	-4%	1.3	-1.3
102	3.2334	3.1137	2.9982	4%	-4%	1.3	-1.3
103	3.1400	3.0229	2.9098	4%	-4%	1.3	-1.3
104	3.0496	2.9350	2.8245	4%	-4%	1.3	-1.3
105	2.9623	2.8502	2.7420	4%	-4%	1.3	-1.3